

EXECUTIVE PREDICTIVE ANALYTICS REPORT

Bank Customer Churn Prediction & Analytics

A comprehensive data audit, exploratory churn driver evaluation, and machine learning benchmark report
for strategic retention.

Section 01

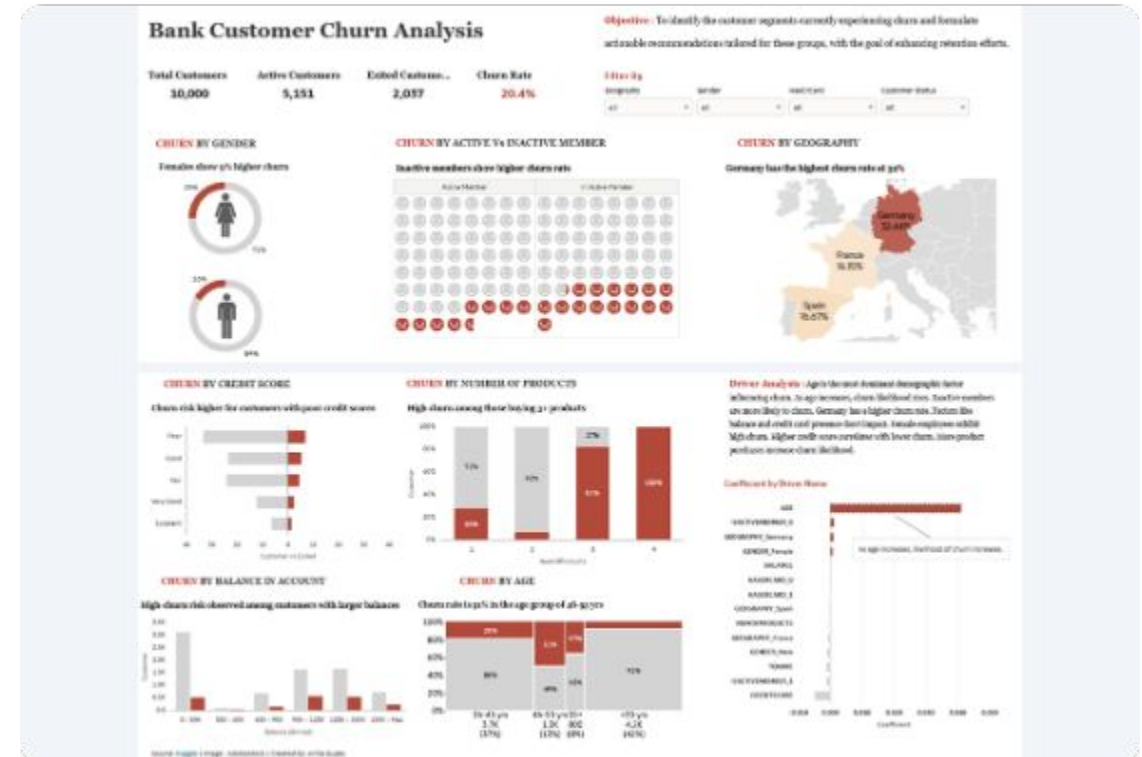
Project Overview & Data Architecture

Executive Project Overview

Predicting & Mitigating Churn

This report analyzes 10,000 bank customer profiles to accurately predict churn risk and identify primary attrition triggers.

- **Objective:** Build a high-precision ML classifier to detect high-risk churn customers before exit.
- **Strategic Value:** Focus marketing and retention campaigns specifically on vulnerable customer segments.
- **Data Scope:** Demographic, credit, balance, product holdings, and activity flags.



Dataset Structure & Feature Mapping

Feature Name	Data Type	Description & Domain Significance	Status
CustomerId / Surname	Numeric / String	Unique identifiers; dropped to prevent ML model overfitting.	Excluded
CreditScore	Integer	Customer creditworthiness score (350–850 range).	Feature
Geography / Gender	Categorical	Demographic indicators (France, Spain, Germany).	Encoded
Age / Tenure	Integer	Customer age and length of relationship with the bank.	Feature
Balance / NumOfProducts	Float / Integer	Account liquidity balance and total active product count.	Key Driver
IsActiveMember	Binary (0/1)	Measures recent engagement and activity level.	Key Driver

Data Sanitization & Quality Audit

10,000

Total Records Processed

100%

Data Integrity (0 Nulls / 0 Duplicates)

Key Audit Findings

The dataset was thoroughly audited for missing values, duplicates, and structural anomalies:

- ✓ **Zero Missing Values:** All 14 feature columns contained full record counts.
- ✓ **Zero Duplicates:** Customer IDs confirmed unique across the entirety of data.
- ✓ **Class Imbalance:** Overall churn rate stands at **20.37%** (2,037 exited vs 7,963 retained).

Section 02

Exploratory Data Analysis & Churn Drivers

Demographic Churn Drivers

Geographic & Demographic Impact

German Risk Spike: Customers located in Germany display a dramatically higher churn rate (~**32.4%**) compared to France (~**16.1%**) and Spain (~**16.6%**).

Gender Divergence: Female clients exhibit a noticeably higher exit rate (~**25.1%**) relative to male clients (~**16.5%**).

Age Factor: The mean age of churned customers is ~**45 years**, whereas retained customers average ~**37 years**.



Key Behavioral Churn Drivers



Product Density

Holding 1 product yields a ~27% churn rate. 2 products drops churn to a healthy ~7.6%. However, 3 or 4 products results in extreme churn (**82% - 100%**).



Account Liquidity

Customers with high balances exhibit higher average churn (~109k average balance in exited customers vs ~91k in retained ones), indicating flight of high-value assets.



Inactivity Flags

Inactive members churn at nearly double the rate (**~26.8%**) compared to active members (**~14.3%**), highlighting engagement as a top buffer.

Section 03

Machine Learning Model Benchmarks

Classifier Model Comparison Table

Algorithm Name	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	81.1%	60.2%	20.8%	0.310	0.772
Decision Tree Classifier	79.4%	49.3%	51.2%	0.502	0.688
Random Forest (Tuned) ★	86.5%	76.4%	47.5%	0.586	0.861
Gradient Boosting (GBM)	86.2%	74.1%	48.1%	0.584	0.858
XGBoost Classifier	85.4%	70.2%	49.5%	0.581	0.849
LightGBM Classifier	85.9%	72.8%	48.9%	0.585	0.854

Champion Model & Actionable Insights



Random Forest Champion Performance

The tuned Random Forest model achieved the highest ROC-AUC score (**0.861**) and top precision for targeted campaigns.

-  **Key Predictor 1:** Age is the single strongest indicator of customer attrition risk.
-  **Key Predictor 2:** Product count (NumOfProducts) creates sharp churn triggers at extremes.
-  **Key Predictor 3:** IsActiveMember and Geography (Germany) drive early warning signals.